

Scientists revive ancient flower

Russian scientists revived a flower which had been frozen for 30,000 years in Siberia
<http://bit.ly/wkHrWB>

Fastest winds ever

NASA's Chandra X-ray Observatory clocked a wind speed of 20 million mph off a black hole which was sucking in the gasses of a nearby star

Motion activated lights

Eliminate the need for switches around the house with this nifty little DIY

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Ever you thought about setting up motion sensors around your home? Imagine all the ugly switchboards disappear, only to be replaced by clean walls. It's one step closer to having a minimalistic theme for your home decor.

The circuit we have this time uses an infrared receiver - transmitter combination to create an invisible line of IR light which when crossed will trigger the light bulb.

Principle

The transmitter is designed for operation over a few meters while not being affected by ambient light. The IR sensor used here is the SFH5110, which has a carrier

frequency of 36 kHz. The IR LED should be capable of switching fast enough to pulsate at 36kHz. This beam is incident on the sensor. When the beam is crossed the receiver triggers the relay which is the switch for whatever you wish to turn on.

The Transmitter

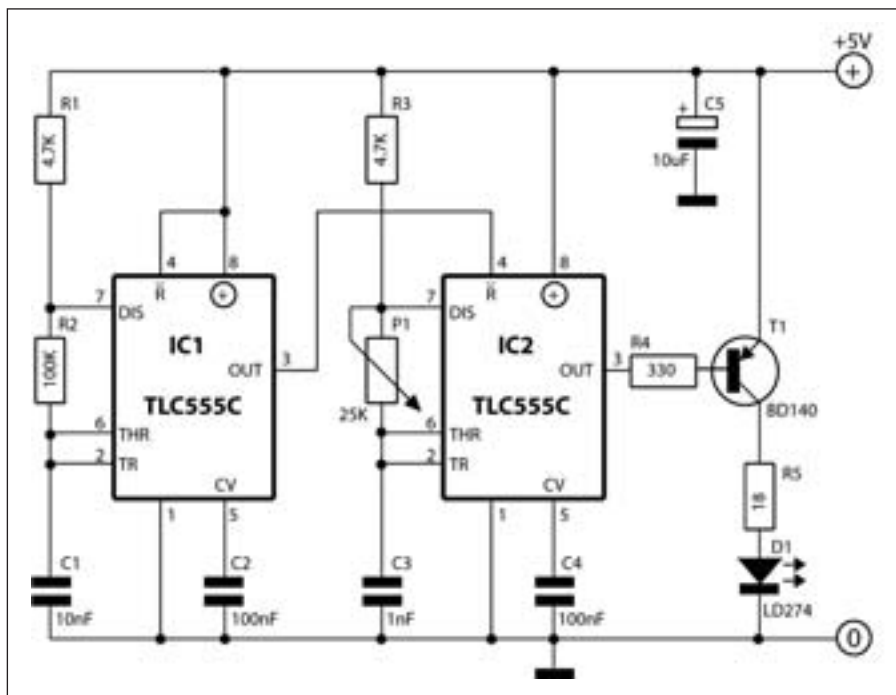
This consists of 2x IC555 timers. We are using the TLC555 as opposed to the NE555 because the NE555 requires a minimum of 4.5 volts whereas the TLC555 is capable of running at 2 volts. Use a voltage regulator (7805) to provide the input supply(+5V). The 555s are configured as Astable Multivibrators or oscillators. The 555 in this configuration follows the equation given at the bottom to generate the pulse signal. The first 555 produces a 300 Hz output to modulate the output of the second 555 which produces the 36kHz signal for the IR LED.

Here R1(taken in ohms) is connected between pins 7 and +5V, R2(taken in ohms) is connected between pins 7 and 6, C1(taken in farads) is connected between

Components list

Parts	Value	Particulars
R1, R3	4.7 K Ω	Resistor
R2	100 K Ω	Resistor
R4	330 Ω	Resistor
R5	18 Ω	Resistor
R6	100 Ω	Resistor
R7, R9	10 K Ω	Resistor
R8	22 K Ω	Resistor
R10	—	Resistor
P1	25 K Ω	Trimmer
C1	10 nF	Polyester Capacitor
C2, C4, C8	100 nF	Polyester Capacitor
C3	1 nF	Polyester Capacitor
C5	10 μ F	Electrolytic Capacitor
C6, C9	47 μ F	Electrolytic Capacitor
C7	1 μ F	Electrolytic Capacitor
IC1, IC2, IC3	TLC555	Timer IC
IC4	MC14043B	Quad R-S Latch
D1	LD274	IR LED
D2	BAT85	Schottky Barrier Diode
D3	1N4148	Diode
T1	BD140	PNP Transistor
T2	BC547	NPN Transistor
IRS	SFH5110	IR Receiver (36kHz)
Relay	—	12V SPST Relay
X	—	AC Light Bulb

2 and ground. f gives the frequency of the output pin 3 in Hertz. For 300Hz we require R1 to be 4.7K, R2 to be about 100-237K and C1 to be 10nF. The output of the first 555(IC1) from pin 3 is fed to pin 4 of the second 555(IC2). Pin 4 is a reset pin which when activated switches the output off. So IC1 switches IC2 on/off about 300 times a second. For IC2, we have R3 at 4.7K, P1 adjusted to 21.6K and C3 at 1nF. The output of IC2 is, hence, a 36kHz signal modulated by 300Hz. Adjust the trimmer preset, P1, to achieve the right frequency. This output is fed to a switching transistor T1(BD140) which turns the IR LED on and off. If you have any problems with using the BD140 which is a PNP transistor, you may replace it with a BD139 which is an NPN transistor. While doing so remember to connect the emitter to the ground and connect the load between



Transmitter circuit diagram

$$f = \frac{1.4}{(R1 + 2R2) \times C1}$$

IC555 astable mode equation